**Algorithm 1** Meta-evolution

**Inputs:** baseline; $G$ generations; $O$ offspring per generation; $P$ population size; tasks $T$;
$\mathcal{S}$ evolvable SR operators (e.g. selection, survival, mutation, loss);
$N_{\text{init}}$ initial seeds; $B$ reevaluation seeds per generation; $N_{\text{val}}$ validation seeds.

```
population ← [baseline]
archive ← {}
for g = 1, ..., G do
    for o = 1, ..., O do                                              ▷ Sample offspring
        operator ← RandomChoice(S)
        mode ← RandomChoice({explore, refine, crossover, simplify})
        parents ← SampleParents(population, mode)
        prompt ← Prompt(T, g, parents, operator, mode)
        x_o ← LLM(prompt)
        Evaluate(x_o, T, N_init)
        archive ← archive ∪ {x_o}
    μ̂_x, n_x ← mean score, seed count of x ∈ archive                  ▷ Reevaluate with TTTS
    ψ_x ← top-two Thompson probability of x under N(μ̂_x, σ̂²/n_x)
    Draw B members from archive with probability ∝ ψ_x
    Evaluate each drawn member on new seeds and update μ̂_x, n_x
    population ← {arg max_{x∈archive} μ̂_x(t) | t ∈ T}                ▷ MAP-Elites, one niche per task
    Fill remaining slots (up to P) by highest overall μ̂_x
for x ∈ population do                                                 ▷ Final validation reevaluation
    val[x] ← Evaluate(x, T_val, N_val)
return arg max_{x∈population} val[x]
```